Installation Manual

Deployment Overview

The project uses a hybrid deployment model. The backend infrastructure is dockerised and started through Docker Compose, while the frontend mobile wallet runs natively through Expo on an iOS simulator, Android emulator, or physical device. This structure was adopted because the backend depends on multiple coordinated services, including the FastAPI application, status manager, Walt.id wallet, issuer and verifier services, and database infrastructure, whereas the mobile frontend requires access to native simulator or device capabilities such as camera scanning. The backend stack includes the application backend, PostgreSQL database, Walt.id wallet API, issuer API, verifier API, verifiable credential repository, and supporting infrastructure services.

System Requirements

To run the system locally, the following software is required:

- Docker and Docker Compose
- Node.js and npm
    - The latest LTS (v24.15.0) installed from nodejs.org
- Expo CLI / Expo development environment
- iOS Simulator, Android Emulator, or Expo Go on a physical mobile device
    - The iOS simulator requires Xcode and the iOS optional packages to be installed – *The iOS simulator does not support QR code scanning as it does not have a camera or emulator for this functionality*
    - For Expo Go on IOS, download the app from the following link: https://apps.apple.com/au/app/expo-go/id982107779
- Python 3.13.5 and PDM, only if running the backend manually without Docker

Internet / Network Requirements

For the app to run as intended, the following is required:

- The mobile app (expo go or simulator) and the desktop computer running the backend are connected to the same network
- The network is a “normal home connection” with a router that provides both devices with a local IP address that can connect to one another
- The network has an internet connection (required for docker compose / QR code generation)

Finding the Local IP Address

On macOS, run:

ipconfig getifaddr en0

On Windows, run:

ipconfig

Use the resulting local network address in both the backend and frontend environment files.

Backend Configuration

Backend Environment Variables

Create a "*.env*" file inside the backend directory. The Docker configuration loads this file into the backend container. The required backend environment variables are:

VERIFIER_PUBLIC_URL=http://YOUR_LOCAL_IP:7003
ISSUER_PUBLIC_URL=http://YOUR_LOCAL_IP:7002

Replace *YOUR_LOCAL_IP* with the local IP address of the machine running the backend. This is required when the frontend is running on a physical device, as localhost will refer to the device itself rather than the development machine.

Backend Setup

From the project root directory, start the backend stack using Docker Compose:

docker compose up –-build

This command starts the FastAPI backend, status manager, Walt.id services, database, and supporting infrastructure. Once running, the main backend API is accessible at:

http://localhost:8000/docs

The standard local service URLs are:

Service	Default Local URL
Issuer Page	http://localhost:8000/issuer
Verifier Page	http://localhost:8000/verifier
Backend API documentation	*http://localhost:8000/docs*
Status Manager API documentation	*http://localhost:8001/docs*
Wallet API	*http://localhost:7001*
Issuer API	*http://localhost:7002*
Verifier API	*http://localhost:7003*

Frontend Configuration

Frontend Setup

In a separate terminal, navigate to the frontend directory:

cd frontend

Install frontend dependencies:

npm install

Create a *".env"* file inside the frontend directory with the following value:

EXPO_PUBLIC_API_BASE_URL=http://YOUR_LOCAL_IP:8000/api

Replace *YOUR_LOCAL_IP* with the local IP address of the machine running the frontend, this allows the Expo application to communicate with the backend API running on the development machine.

Running the Frontend

Start the Expo development server from the frontend directory:

npm start

Once Expo has started, run the application using one of the following options:

- Press 'i' to launch the iOS simulator
- Press 'a' to launch the Android emulator
- Press 's' to switch to Expo Go build
- Scan the Expo QR code using phone camera on iPhone or Expo Go app on Android

When using a physical device, the phone and development machine must be connected to the same Wi-Fi network

End-to-End Usage Flow

Once both backend and frontend are running:

1. Open the issuer interface at http://localhost:8000/issuer
2. Generate a credential offer
3. Scan the QR code using the wallet application
4. Store the credential locally within the wallet
5. Open the verifier interface at http://localhost:8000/verifier
6. Generate a presentation request
7. Scan the request QR code using the wallet

8. Select credentials and complete the presentation

This process can be customized using a range of credential structures found within the issuer and verifier page to reflect a variety of real-world processes.

Testing

Frontend tests can be run from the frontend directory using:

npm test

Coverage can be generated using:

npm run test:coverage

Backend tests are executed using *pytest* from the backend directory, this also shows test coverage:

STATUS_MANAGER_API_BASE_URL=http://localhost:8001 WALLET_API_BASE_URL=http://localhost:7001 pdm run pytest tests/ --cov=. --cov-report=term-missing --cov-fail-under=80 -v

Test coverage is enforced with a minimum threshold of 80%, and additional validation is performed through a continuous integration pipeline, which runs backend tests, linting, and Docker build checks on each repository update.